

7.2.09. Let $X = \{1, 2, 3\}$, $Y = \{1, 2, 3, 4\}$, and $Z = \{1, 2\}$.

- a. Define a function $f : X \rightarrow Y$ that is injective but not surjective.
- b. Define a function $g : X \rightarrow Z$ that is surjective but not injective
- c. Define a function $h : X \rightarrow X$ that is neither injective or surjective.
- d. Define a function $k : X \rightarrow X$ that is injective and surjective but is not the identity function on X .

- a. Define a function $f : X \rightarrow Y$ that is injective but not surjective.

Let $X = \{1, 2, 3\}$

Let $Y = \{1, 2, 3, 4\}$

Definition of injective:

Outputs cannot be shared

Hence, $f : f(x) = x$

$$f(1) = 1$$

$$f(2) = 2$$

$$f(3) = 3$$

Solution: $f : f(x) = x$

- b. Define a function $g : X \rightarrow Z$ that is surjective but not injective

Let $X = \{1, 2, 3\}$

Let $Z = \{1, 2\}$

Definition of surjective:

Outputs cannot go unused

Hence, $g : x \text{ is odd} = 1, x \text{ is even} = 2$

$$g(1) = 1$$

$$g(2) = 2$$

$$g(3) = 1$$

Solution: $g : x \text{ is odd} = 1, x \text{ is even} = 2$

- c. Define a function $h : X \rightarrow X$ that is neither injective or surjective.

Let $X = \{1, 2, 3\}$

Definition of injective negated:

Outputs can be shared

Definition of surjective negated:

Outputs can go unused

Hence, $h : h(x) = 1$

$$h(1) = 1$$

$$h(2) = 1$$

$$h(3) = 1$$

Solution: $h : h(x) = 1$

- d. Define a function $k : X \rightarrow X$ that is injective and surjective but is not the identity function on X .

Let $X = \{1, 2, 3\}$

Definition of injective:

Outputs cannot be shared

Definition of surjective:

Outputs cannot go unused

Since $k \neq$ the identity function on X , then:

$$k : x < 2 \rightarrow 3 \text{ and } x \geq 2 \rightarrow x - 1$$

$$k(1) = 3$$

$$k(2) = 1$$

$$k(3) = 2$$

Solution: $k : x < 2 \rightarrow 3 \text{ and } x \geq 2 \rightarrow x - 1$